

ADAM ZAATAR

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EDUCATION

Bowdoin College

Brunswick, ME

B.A., Computer Science and Economics, Minor in German

Expected May 2027 · GPA 3.53

Relevant coursework: Operating Systems, Computer Systems, Data Structures & Algorithms, Artificial Intelligence, Econometrics, Cybersecurity Learning Assistant, CSCI 1101 (Intro to CS), Bowdoin CS Department — Fall 2026

EXPERIENCE

Software Engineering Intern, ProgressSoft Corporation

Amman, Jordan · Jun–Aug 2026

- Built EventGuard, a seven-module Java 21 and Spring Boot payment-file import and audit application: NIO file reading, CSV parsing, duplicate payment-ID detection, and typed row-level rejection handling that separates parser errors from business-rule rejections.
- Structured the system as ports and adapters, keeping the core domain free of framework dependencies, pointing every adapter at core ports rather than the reverse, and assembling concrete implementations only at the CLI and Spring composition roots.
- Implemented transactional JDBC persistence to PostgreSQL: explicit commit and rollback with suppressed-exception handling, try-with-resources on every connection, statement, and result set, parent import-attempt rows with generated keys and child payment rows, and BigDecimal against a NUMERIC column to avoid floating-point drift.
- Added a Spring MVC multipart upload endpoint with filename sanitization against path traversal, temporary-file cleanup that preserves the original failure, and structured HTTP error mapping; verified the system with 107 passing tests, including Spring Boot slices and a live JDBC integration test.

PROJECTS

Operating Systems (CSCI 3310)

C / C++

- Built a user-level thread library: ucontext context switching, FIFO scheduling, locks and condition variables under Mesa semantics, and atomicity through interrupt masking, validated with a test suite written to expose buggy implementations.
- Implemented an external virtual-memory pager: demand paging, second-chance (clock) page replacement, software-tracked reference and dirty bits, and disk swap, with work deferral tuned to the cost gap between faults, zero-fills, and disk I/O.

ClickErase

Python, Gradio, MobileSAM, LaMa, Hugging Face Spaces

- Built and deployed a public image-editing app: click an object, generate a segmentation mask with MobileSAM, and remove it with LaMa inpainting. Live demo and github.com/adamzatar/clickerase.

RESEARCH

- AI Layoffs and Investor Reactions (advised by Prof. Botsch): hand-coded events, cross-checked SEC filings, and estimated abnormal returns via regression. Gibbons Summer Research Fellowship (funded, advised by Prof. Erik Nelson): econometric modeling and experimental design.

SKILLS

Languages: Java, Python, C, C++, SQL, JavaScript/TypeScript

Backend: Spring Boot, Spring MVC, REST APIs, JDBC, Maven

Testing: JUnit 5, Mockito, test-driven development

Systems: Linux/POSIX, multithreading, concurrency, virtual memory

Databases: PostgreSQL, MySQL, SQLite, Redis

Spoken Languages: Arabic (native), English (fluent), German (intermediate)

LEADERSHIP

- Founder and President, Bowdoin Martial Arts Club (secured \$800 in funding, grew membership to over 30); Student Advisor, Bowdoin CS Department faculty hiring.